

Labor Income Taxes and Transfers: Optimal Policy Theory

Follows loosely Chapters 20-21 of Gruber

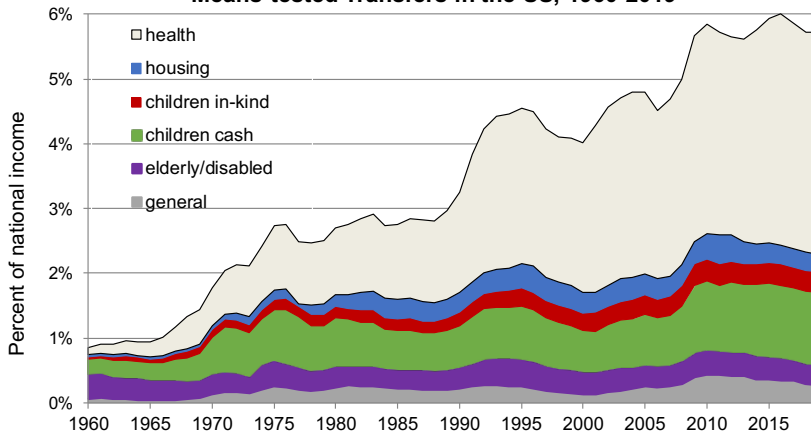
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- **Key question:** Should government reduce inequality using taxes and transfers?
- Governments use **taxes** to raise revenue
- This revenue funds **transfer** programs:
 - Universal Transfers: Public Education, Health Care Benefits (only 65+ in the US), Retirement and Disability Benefits, Unemployment benefits
 - Means-tested Transfers: In-kind (Medicaid, public housing, foodstamps in the US) and cash benefits
- Modern governments raise large fraction of GDP in taxes (30-45%) and spend significant fraction of GDP on transfers

References: Comprehensive description in: <http://www.taxpolicycenter.org/taxfacts/>

- **A) Taxes:** (1) individual income tax (fed+state), (2) payroll taxes on earnings (fed, funds Social Security+Medicare), (3) corporate income tax (fed+state), (4) sales taxes (state)+excise taxes (state+fed), (5) property taxes (state)
- **B) Means-tested Transfers:** (1) refundable tax credits (fed), (2) in-kind transfers (fed+state): Medicaid, public housing, nutrition (SNAP), education, (3) cash welfare: TANF for single parents (fed+state), SSI for old/disabled (fed)

Means-tested Transfers in the US, 1960-2019



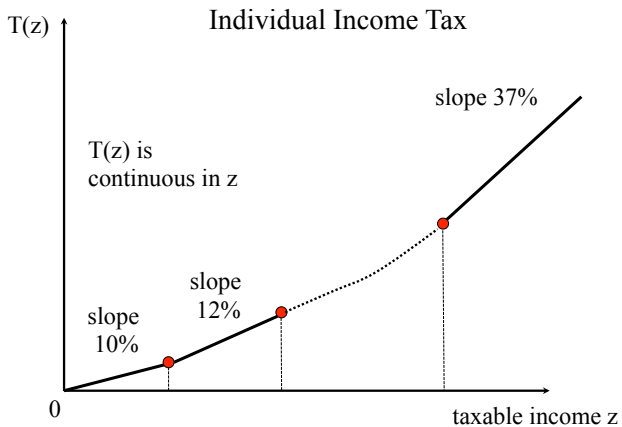
Source. National Accounts. Includes all individualized and means-tested transfers. General is untarggetted (SNAP and general assistance for adults). Children cash includes refundable tax credits (EITC+CTC), TANF, and SNAP for children. Health is mostly Medicaid.

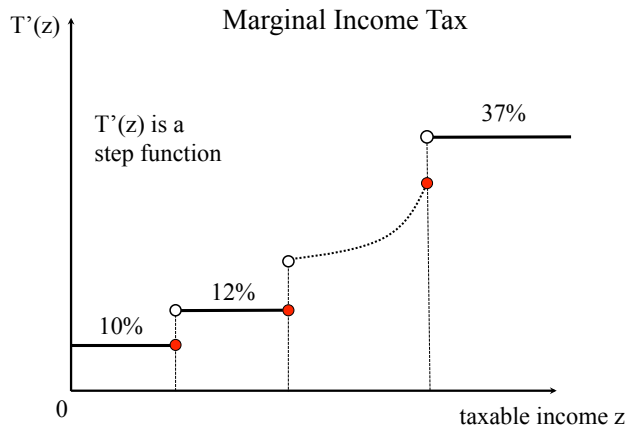
- US income tax assessed on **annual family** income (not individual) [most other OECD countries have shifted to individual assessment]
- Sum all cash income sources from family members (both from labor and capital income sources) = called **Adjusted Gross Income (AGI)**
- Main exclusions: fringe benefits (health insurance, pension contributions and returns), imputed rent of homeowners, undistributed corporate profits, unrealized capital gains
- $\Rightarrow$ AGI base is only 70% of national income

- Taxable income = AGI - deductions
- Deduction is max of standard deduction or itemized deductions
- Standard deduction is a fixed amount (\$12K for singles, \$24K for married couple)
- Itemized deductions: (a) state and local taxes paid (up to \$10K), (b) mortgage interest payments, (c) charitable giving, various small other items
- About 10% of AGI lost through itemized deductions, called tax expenditures

Federal US Income Tax: Tax Brackets

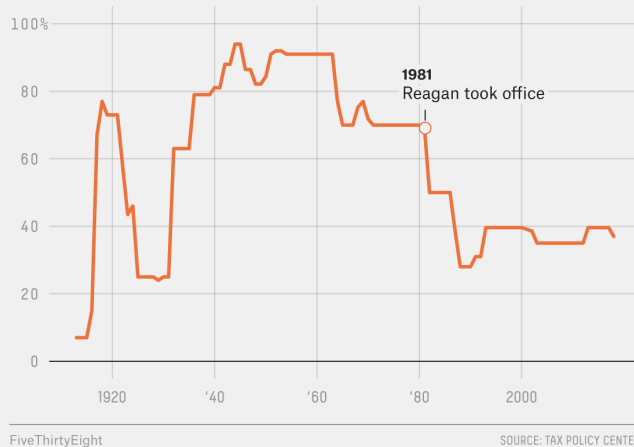
- Tax $T(z)$ is **piecewise linear** and **continuous** function of taxable income z with constant marginal tax rates (MTR) $T'(z)$ by brackets [see graph]
- In 2018+, 6 brackets with MTR 10%, 12%, 22%, 24%, 32%, 35%, 37% (top bracket for z above \$600K), indexed on price inflation
- Lower preferential rates (up to a max of 20%) apply to dividends (since 2003) and realized capital gains [in part to offset double taxation of corporate profits]
- Tax rates change frequently over time. Top MTRs have declined drastically since 1960s (as in most OECD countries)





Historically, a 70 percent marginal tax rate is not unusual

The top marginal income tax rates from 1913 to 2018

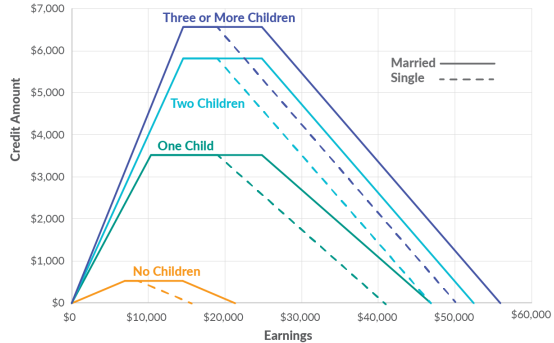


Federal US Income Tax: Tax Credits

- **Tax credits:** Additional reduction in taxes
- (1) **Non refundable** (cannot reduce taxes below zero): foreign tax credit, child care expenses, education credits, energy credits, and many others
- (2) **Refundable** (can reduce taxes below zero, i.e., be net transfers): EITC (earned income tax credit, up to \$3.5K, \$5.7K, \$6.5K for working families with 1, 2, 3+ kids), Child Tax Credit (\$2K per kid, partly refundable)
- Refundable tax credits are now the largest means-tested cash transfer for low income families

The Phase-In and Phaseout of the EITC

Credit Amount by Marital Status and Number of Children

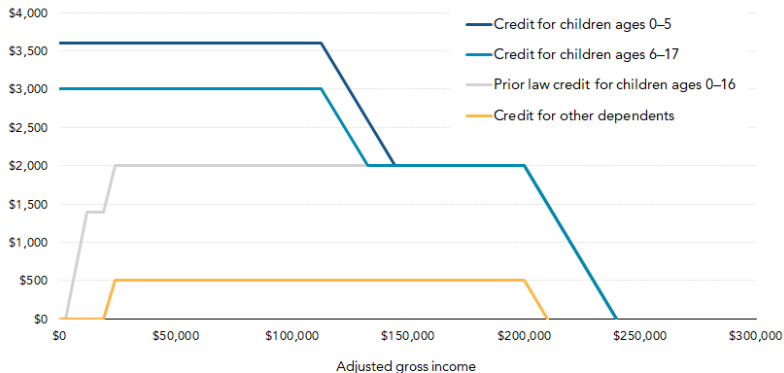


Source: Amir El-Sibaie, "2019 Tax Brackets," Tax Foundation, Nov. 28, 2018.

TAX FOUNDATION

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FIGURE 1
Child Tax Credit, Single Parent
For one child, tax year 2021



Source: Urban-Brookings Tax Policy Center calculations.

Notes: Assumes all income comes from earnings, and child meets all tests to be a CTC-qualifying dependent. \$3,000 and \$3,600 credits are fully refundable; prior law limited refunds to \$1,400 out of the maximum \$2,000 credit. Credit for married parents first phases out at \$150,000 of income until credit reaches pre-2021 level; begins second phase out at \$400,000 of income. Only citizen children qualify for the \$3,000 and \$3,600 credits for children under 18. Noncitizens under age 18 who meet the dependency tests of eligibility can qualify for other dependent credit.

Federal US Income Tax: Tax Filing

- Taxes on year t earnings are withheld on paychecks during year t (pay-as-you-earn)
- Income tax return filed in Feb-April 15, year $t + 1$ [filers use either software or tax preparers, huge private industry, most OECD countries provide pre-populated returns]
- Most tax filers get a tax refund as withholdings larger than taxes owed in general
- Payers (employers, banks, etc.) send income information to govt (3rd party reporting)
- 3rd party reporting + withholding at source is key for successful enforcement

Main Means-tested Transfer Programs

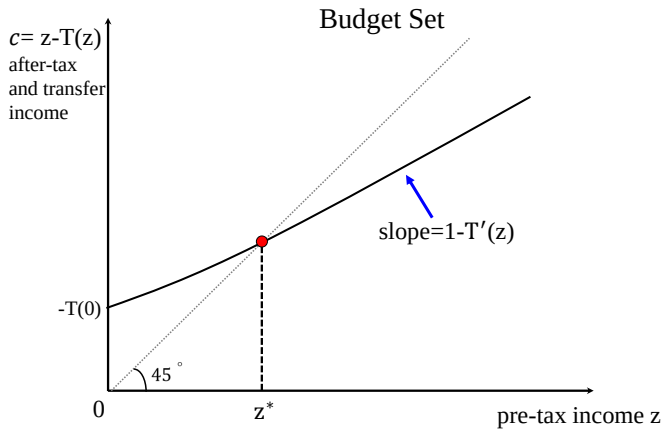
- ① **Traditional transfers:** managed by welfare agencies, paid on monthly basis, high stigma and take-up costs $\Rightarrow$ low take-up rates (often only around 50%)
 - Main programs: Medicaid (health insurance for low incomes), SNAP (former food stamps), public housing, TANF (welfare), SSI (aged+disabled)
- ② **Refundable income tax credits:** managed by tax administration, paid as an annual lumpsum in year $t + 1$, low stigma and take-up cost $\Rightarrow$ high take-up rates
 - Main programs: EITC and Child Tax Credit [large expansion since the 1990s] for low income working families with children

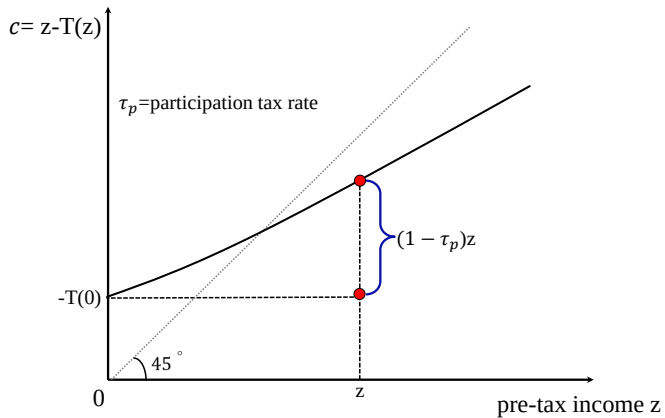
Key Concepts for Taxes/Transfers

- Budget set in graphs $(z, z - T(z))$ which integrates taxes and transfers
- 1) Transfer benefit with zero earnings $-T(0)$ [sometimes called demogrant or lumpsum grant]
- 2) Marginal tax rate (or phasing-out rate) $T'(z)$: individual keeps $1 - T'(z)$ for an additional \$1 of earnings (intensive labor supply response)
- 3) Participation tax rate $\tau_p = [T(z) - T(0)]/z$: individual keeps fraction $1 - \tau_p$ of earnings when moving from zero earnings to earnings z (extensive labor supply response):

$$z - T(z) = -T(0) + z \cdot (1 - \tau_p)$$

- 4) Break-even earnings point z^* : point at which $T(z^*) = 0$

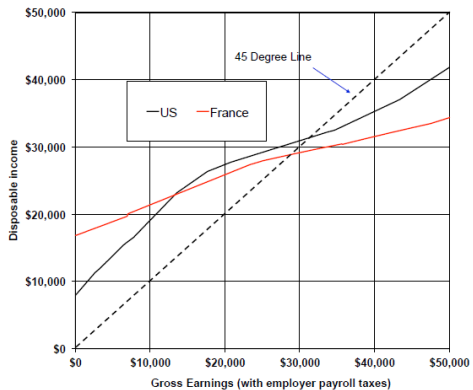




US Tax/Transfer System, single parent with 2 children, 2009



Source: Computations made by Emmanuel Saez using tax and transfer system parameters



Source: Piketty, Thomas, and Emmanuel Saez (2012)

France provides larger transfers to zero-earners than does the United States.

- A. True
- B. False
- C. Uncertain

Profile of Current Means-tested Transfers

- Traditional means-tested programs reduce incentives to work for low income workers
- Refundable tax credits have significantly increased incentive to work for low income workers
- However, refundable tax credits cannot benefit those with zero earnings
- Trade-off: US chooses to reward work more than most European countries (such as France) but therefore provides smaller benefits to those with no earnings

Optimal Taxation: Case with No Behavioral Responses

- Utility $u(c)$ strictly increasing and concave
- Same for everybody where c is after tax income.
- Income z is fixed for each individual, $c = z - T(z)$ where $T(z)$ is tax/transfer on z (tax if $T(z) > 0$, transfer if $T(z) < 0$)
- N individuals with fixed incomes $z_1 < \dots < z_N$
- Government maximizes **Utilitarian** objective: $SWF = \sum_{i=1}^N u(z_i - T(z_i))$ subject to **budget constraint** $\sum_{i=1}^N T(z_i) = 0$ (taxes need to fund transfers)

Simple Model With No Behavioral Responses

- Replace $T(z_1) = -\sum_{i=2}^N T(z_i)$ from budget constraint
- First order condition (FOC) in $T(z_i)$ for a given $i = 2, \dots, N$:

$$0 = \frac{\partial SWF}{\partial T(z_i)} = u' \left(z_1 + \sum_{j=2}^N T(z_j) \right) - u'(z_i - T(z_i)) = 0 \Rightarrow$$

- $u'(z_i - T(z_i)) = u'(z_1 - T(z_1)) \Rightarrow z_i - T(z_i) = \text{constant across } i = 1, \dots, N$
- Perfect equalization of after-tax income = 100% tax rate and redistribution [see graph]
- Utilitarianism with decreasing marginal utility leads to perfect egalitarianism [Edgeworth, 1897]

Simpler Derivation with Just Two Individuals

$$\max SWF = u(z_1 - T(z_1)) + u(z_2 - T(z_2)) \text{ s.t. } T(z_1) + T(z_2) = 0$$

Replace $T(z_1) = -T(z_2)$ in SWF using budget constraint:

$$SWF = u(z_1 + T(z_2)) + u(z_2 - T(z_2))$$

First order condition (FOC) in $T(z_2)$:

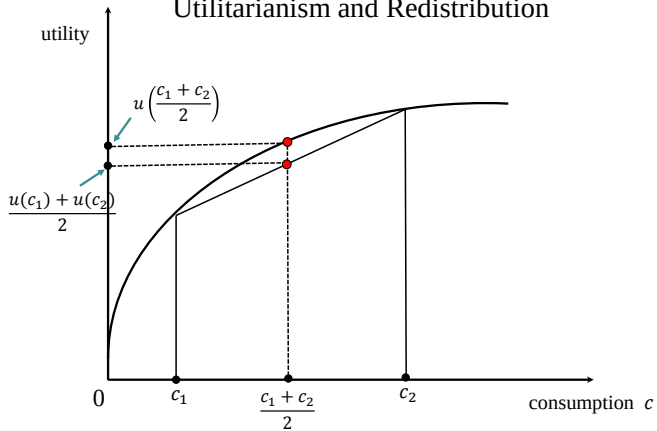
$$0 = \frac{dSWF}{dT(z_2)} = u'(z_1 + T(z_2)) - u'(z_2 - T(z_2)) = 0 \Rightarrow$$

$$u'(z_1 + T(z_2)) = u'(z_2 - T(z_2)) \Rightarrow u'(z_1 - T(z_1)) = u'(z_2 - T(z_2))$$

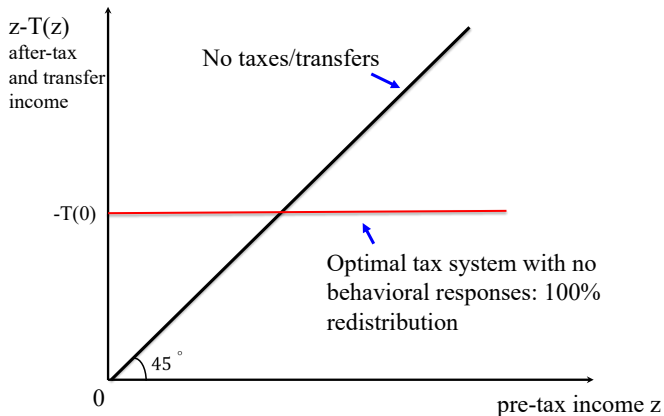
$\Rightarrow z_1 - T(z_1) = z_2 - T(z_2)$ constant across the 2 individuals

Perfect equalization of after-tax income = 100% tax rate and redistribution [see graph]

Utilitarianism and Redistribution



Optimal Tax/Transfer Systems



- ① **No behavioral responses:** Obvious missing piece: 100% redistribution would destroy incentives to work and thus the assumption that z is exogenous is unrealistic

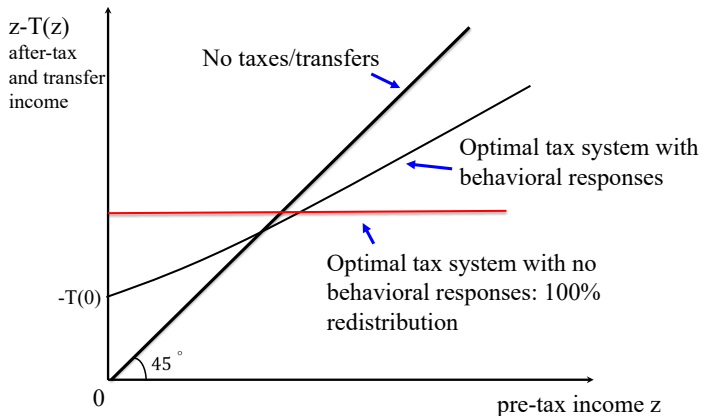
⇒ Optimal income tax theory incorporates behavioral responses
- ② **Issue with Utilitarianism:** Even absent behavioral responses, many people would object to 100% redistribution [perceived as confiscatory]

⇒ Citizens' views on fairness impose **bounds** on redistribution govt can do [political economy / public choice theory] (see e.g. Weinzierl 2014)

Equity-efficiency Trade-off

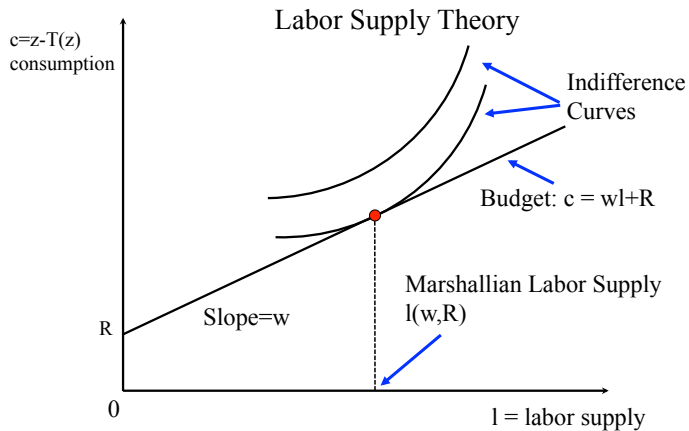
- Taxes can be used to raise revenue for transfer programs which can reduce inequality in disposable income $\Rightarrow$ Desirable if society feels that inequality is too large
- Taxes (and transfers) reduce incentives to work $\Rightarrow$ High tax rates create economic inefficiency if individual respond to taxes
- Size of behavioral response limits the ability of govt to redistribute with taxes/transfers
 $\Rightarrow$ Generates an equity-efficiency trade-off
- Empirical tax literature estimates the size of behavioral responses to taxation

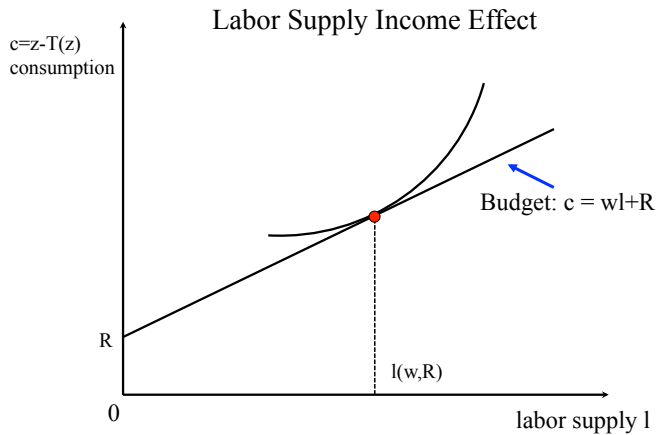
Optimal Tax/Transfer Systems

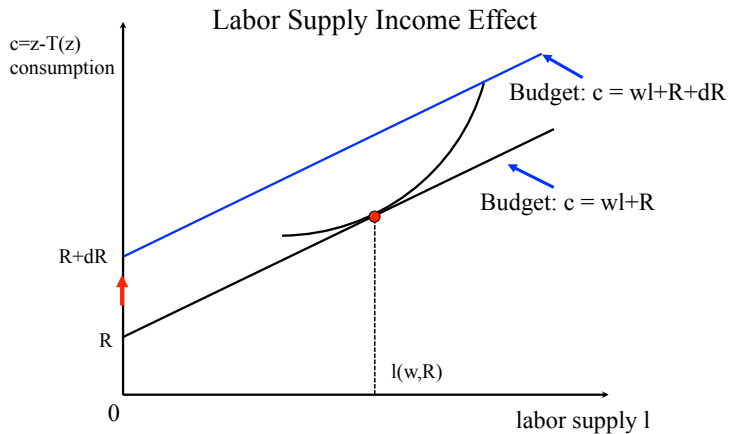


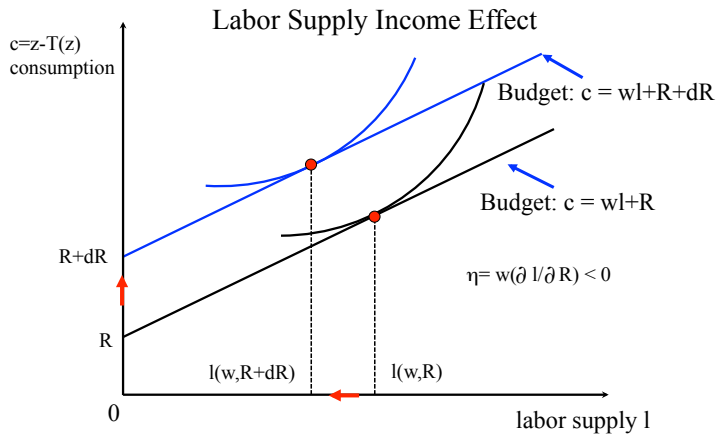
Labor Supply Theory

- Individual has utility over labor supply l and consumption c : $u(c, l)$ increasing in c and decreasing in l [i.e., increasing in leisure]
- $\max u(c, l)$ subject to $c = w \cdot l + R$ with $w = \bar{w} \cdot (1 - \tau)$ the net-of-tax wage ($\bar{w}$ is before tax wage rate and τ is tax rate), and R non-labor income
- FOC $w \frac{\partial u}{\partial c} + \frac{\partial u}{\partial l} = 0$ defines Marshallian labor supply $l = l(w, R)$
- Uncompensated labor supply elasticity: $\varepsilon^u = \frac{w}{l} \cdot \frac{\partial l}{\partial w}$
- Income effects: $\eta = l \frac{\partial l}{\partial R} \leq 0$ (if leisure is a normal good)







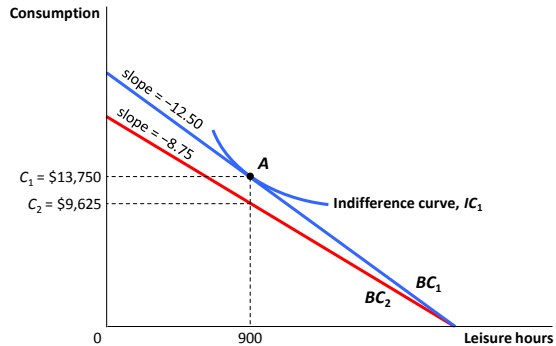


Labor Supply Theory

- Substitution effects: Hicksian labor supply: $I^c(w, u)$ minimizes cost needed to reach u given slope $w \Rightarrow$
Compensated elasticity: $\varepsilon^c = \frac{w}{I} \cdot \frac{\partial I^c}{\partial w} > 0$
- Slutsky equation: $\frac{\partial I}{\partial w} = \frac{\partial I^c}{\partial w} + I \frac{\partial I}{\partial R} \Rightarrow \varepsilon^u = \varepsilon^c + \eta$
- Tax rate τ discourages work through substitution effects (work pays less at the margin)
- Tax rate τ encourages work through income effects (taxes make you poorer and hence in more need of income)
- Net effect ambiguous (captured by sign of ε^u)

21.1

Basic Theory

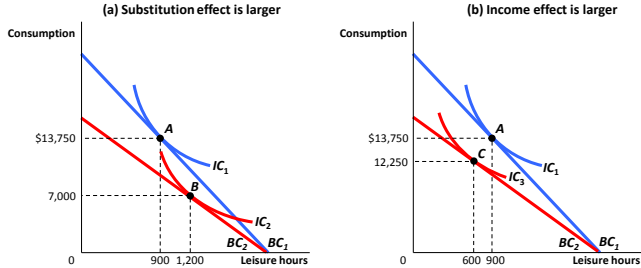


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5 of 29

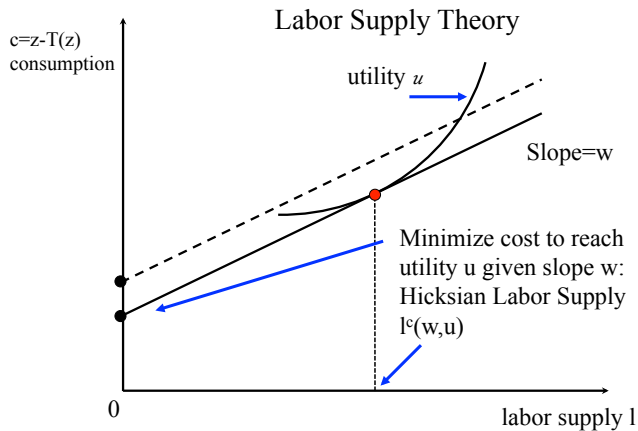
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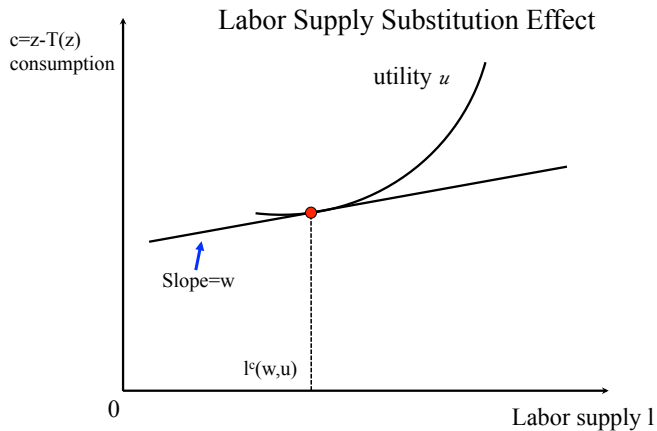
Substitution versus Income Effect

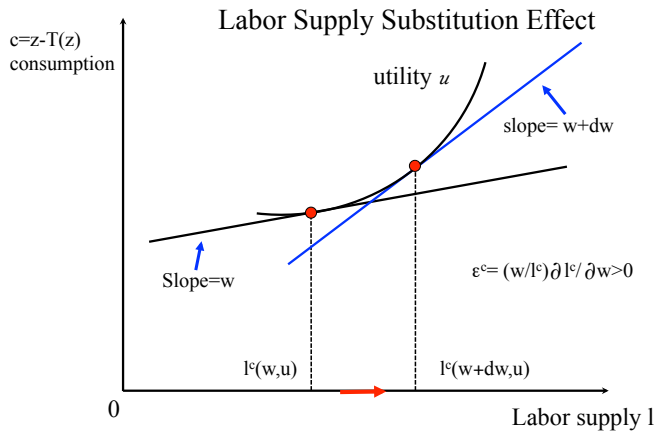


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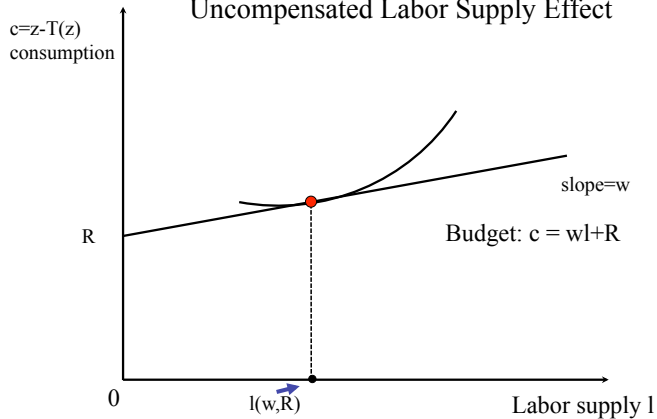
7 of 29

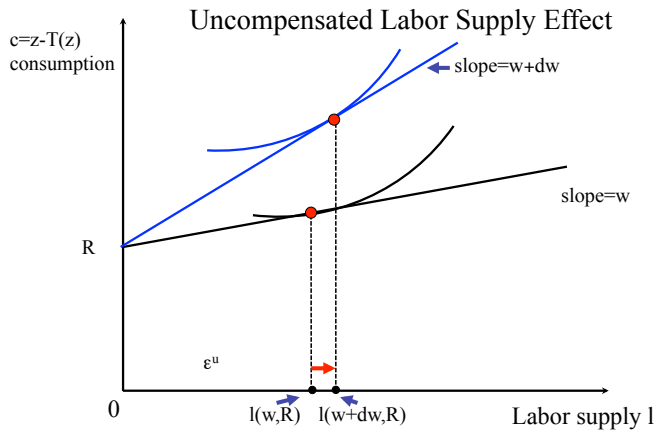


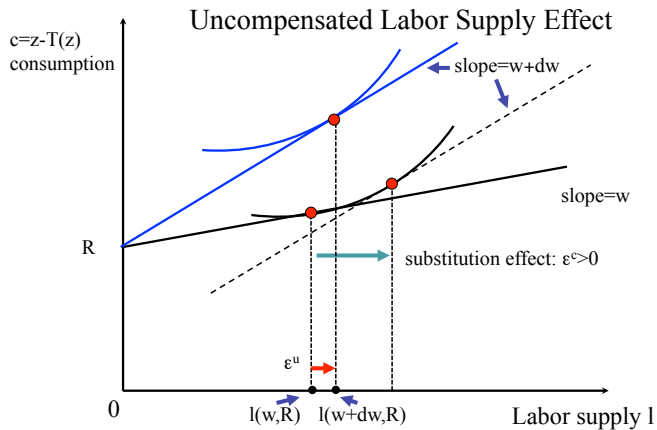


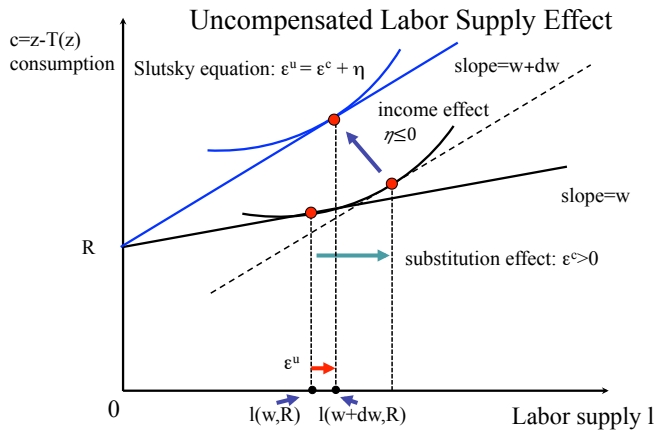


Uncompensated Labor Supply Effect





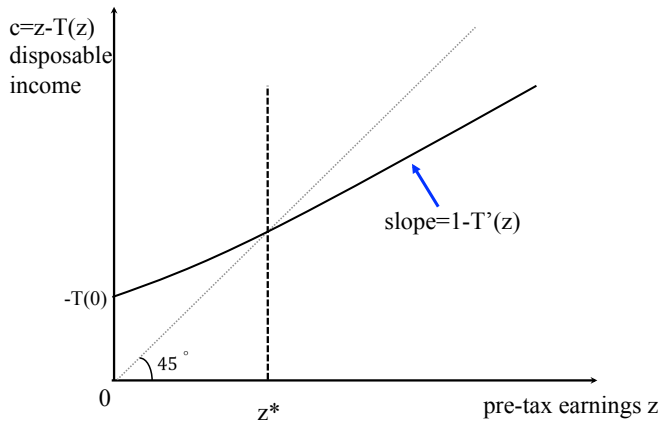




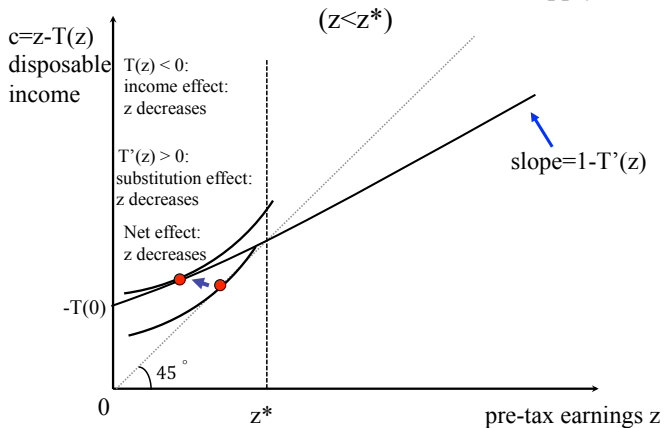
General nonlinear income tax [see graph]

- With no taxes: $c = z$ (consumption = earnings)
- With taxes $c = z - T(z)$ (consumption = earnings - net taxes)
- $T(z) \geq 0$ if individual pays taxes on net, $T(z) \leq 0$ if individual receives transfers on net
- $T'(z) > 0$ reduces net wage rate and reduces labor supply through substitution effects
- $T(z) > 0$ reduces disposable income and increases labor supply through income effects
- $T(z) < 0$ increases disposable income and decreases labor supply through income effects
- Transfer program such that $T(z) < 0$ and $T'(z) \geq 0$ always discourages labor supply

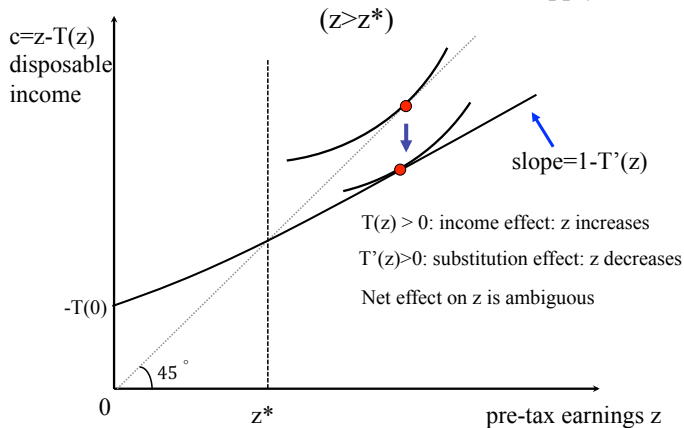
Effect of Taxes/Transfers on Labor Supply



Effect of Taxes/Transfers on Labor Supply



Effect of Taxes/Transfers on Labor Supply

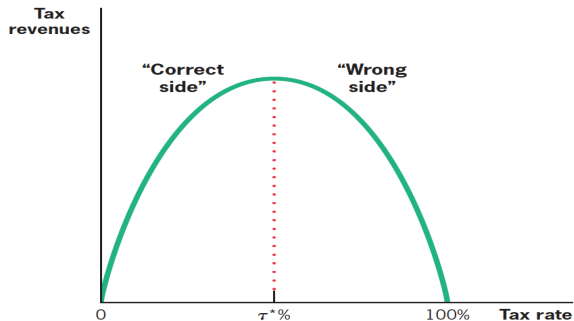


Optimal Linear Tax Rate: Laffer Curve

- $c = (1 - \tau) \cdot z + R$ with τ linear tax rate and R fixed universal transfer funded by taxes
 $R = \tau Z$ with Z average earnings
- Individual i choose l_i to maximize $u^i((1 - \tau) \cdot w_i l_i + R, l_i)$
- Labor supply choices l_i determine individual earnings $z_i = w_i l_i$. Assume small income effects $\Rightarrow$ average earnings $Z = \sum_i z_i / N$ depend positively on net-of-tax rate $1 - \tau$.
- Tax Revenue per person $R(\tau) = \tau \cdot Z(1 - \tau)$ is inversely U-shaped with τ : $R(\tau = 0) = 0$ (no taxes) and $R(\tau = 1) = 0$ (nobody works): called the Laffer Curve

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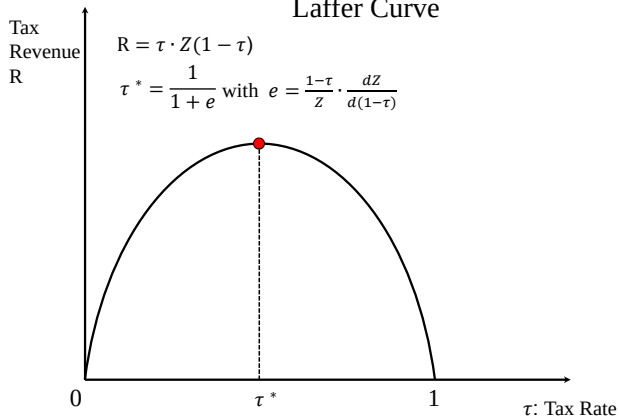
The Laffer Curve



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31 of 40

Laffer Curve



Optimal Linear Tax Rate: Laffer Curve

- Top of the Laffer Curve is at τ^* maximizing tax revenue:

$$0 = R'(\tau^*) = Z - \tau^* \frac{dZ}{d(1-\tau)} \Rightarrow \frac{\tau^*}{1-\tau^*} \cdot \frac{1-\tau^*}{Z} \frac{dZ}{d(1-\tau)} = 1$$

- Revenue maximizing tax rate: $\tau^* = \frac{1}{1+e}$ with $e = \frac{1-\tau}{Z} \frac{dZ}{d(1-\tau)}$ is the elasticity of average income Z with respect to the net-of-tax rate $1-\tau$ [empirically estimable]
- Inefficient to have $\tau > \tau^*$ because decreasing τ would make taxpayers better off (they pay less taxes) and would increase tax revenue for the government [and hence univ. transfer R]
- If government is **Rawlsian** (maximizes welfare of the worst-off person with no earnings) then $\tau^* = 1/(1+e)$ is optimal to make transfer $R(\tau)$ as large as possible

Optimal Linear Tax Rate: Formula

Government chooses τ to maximize **utilitarian** social welfare

$$SWF = \sum_i u^i((1 - \tau)w_i l_i + \tau \cdot Z(1 - \tau), l_i)$$

taking into account that labor supply l_i responds to taxation and hence that this affects the tax revenue per person $\tau \cdot Z(1 - \tau)$ that is redistributed back as transfer to everybody

Government first order condition: (using the envelope theorem as l_i maximizes u^i):

$$0 = \frac{dSWF}{d\tau} = \sum_i \frac{\partial u^i}{\partial c} \cdot \left[-z_i + Z - \tau \frac{dZ}{d(1 - \tau)} \right]$$

Optimal Linear Tax Rate: Formula

- Hence, we have the following optimal linear income tax formula

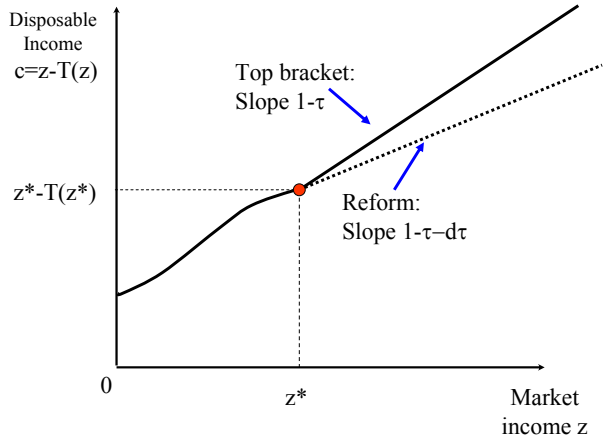
$$\tau = \frac{1 - \bar{g}}{1 - \bar{g} + e} \quad \text{with} \quad \bar{g} = \frac{\sum_i z_i \cdot \frac{\partial u^i}{\partial c}}{Z \cdot \sum_i \frac{\partial u^i}{\partial c}}$$

- $0 \leq \bar{g} < 1$ as $\frac{\partial u^i}{\partial c}$ is decreasing with z_i (marginal utility falls with consumption)
- τ decreases with elasticity e [efficiency] and with parameter $\bar{g}$ [equity]
- Formula captures the **equity-efficiency trade-off**
- $\bar{g}$ is low and τ close to Laffer rate $\tau^* = 1/(1 + e)$ when (a) inequality is high, (b) marginal utility decreases fast with income

Optimal Top Income Tax Rate (Diamond and Saez JEP'11)

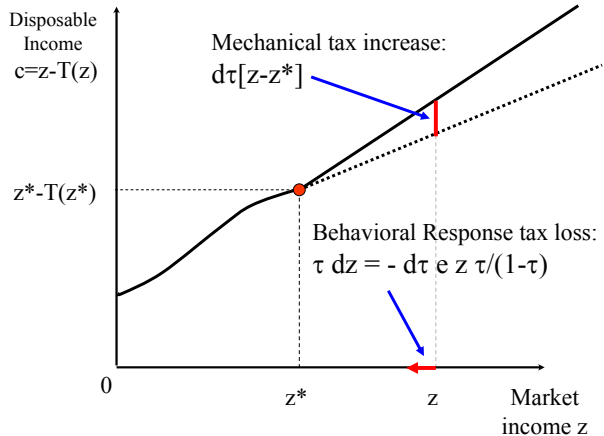
- In practice, individual income tax is progressive with brackets with increasing marginal tax rates. What is the optimal top tax rate?
- Consider constant MTR τ above fixed z^* . Goal is to derive optimal τ
- In the US in 2020, $\tau = 39\%$ and $z^* = 622,050$ (married filing jointly) ($\simeq$ top 1%)
- Denote by z average income of top bracket earners [depends on net-of-tax rate $1 - \tau$], with elasticity $e = [(1 - \tau)/z] \cdot dz/d(1 - \tau)$
- Suppose the government wants to maximize tax revenue collected from top bracket taxpayers (marginal utility of consumption of top 1% earners is small)

Optimal Top Income Tax Rate (Mirrlees '71 model)



Source: Diamond and Saez JEP'11

Optimal Top Income Tax Rate (Mirrlees '71 model)



Source: Diamond and Saez JEP'11

Optimal Top Income Tax Rate

- Consider small $d\tau > 0$ reform above z^* . Denote mean income above z^* as z_m . Assume constant elasticity e above z^* .
- 1) **Mechanical increase** in tax revenue: $dM = [z_m - z^*]d\tau$
- 2) **Behavioral response** reduces tax revenue:
$$dB = \tau dz_m = -\tau \frac{dz_m}{d(1-\tau)} d\tau = -\frac{\tau}{1-\tau} \cdot e \cdot z_m \cdot d\tau$$
- $dM + dB = d\tau \left\{ [z_m - z^*] - e \frac{\tau}{1-\tau} z_m \right\}$
- Optimal τ such that $dM + dB = 0$: $\frac{\tau}{1-\tau} = \frac{1}{e} \cdot \frac{z_m - z^*}{z_m} \Rightarrow \tau = \frac{1}{1+a \cdot e}$ with $a = \frac{z_m}{z_m - z^*}$

Optimal Top Income Tax Rate

Optimal top tax rate: $\tau = \frac{1}{1+a \cdot e}$ with $a = \frac{z_m}{z_m - z^*}$

- Optimal τ decreases with e [efficiency]
- Optimal τ decrease with a [thinness of top tail]
- Empirically $a \simeq 1.5$, easy to estimate using distributional data [mean income above $z^* = \$0.5\text{m}$ is about $\$1.5\text{m}$ in the US]
- Empirically e is harder to estimate [controversial]
- **Examples:**
 - If $e = .5$ then $\tau = 1/(1 + 1.5 \cdot 0.5) = 1/1.75 = 57\%$
 - If $e = .25$ then $\tau = 1/(1 + 1.5 \cdot 0.25) = 1/1.375 = 73\%$

Real vs. Tax Avoidance Responses

- Behavioral response to income tax comes not only from reduced labor supply but from tax avoidance or tax evasion
- Tax avoidance: legal means to reduce tax liability (exploiting tax loopholes)
- Tax evasion: illegal under-reporting of income
- Labor supply vs. tax avoidance/evasion distinction matters because:
 - If people work less when tax rates increase, there is not much the government can do about it
 - If people avoid/evade more when tax rates increase, then the govt can reduce tax avoidance/evasion opportunities [closing tax loopholes, broadening the tax base, increasing tax enforcement, etc.]

Real vs. Avoidance Responses

- **Key policy question:** Is it possible to eliminate avoidance responses using base broadening, etc.? or would new avoidance schemes keep popping up?
- a) Some forms of tax avoidance are due to **poorly designed tax codes** (preferential treatment for some income forms or some deductions)
- b) Some forms of tax avoidance/evasion can only be addressed with **international cooperation** (off-shore tax evasion in tax heavens)
- c) Some forms of tax avoidance/evasion are due to technological limitations of tax collection (impossible to tax informal cash businesses)

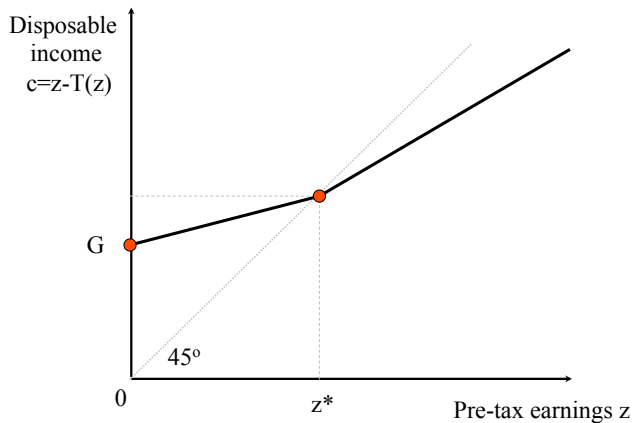
Optimal Profile of Transfers

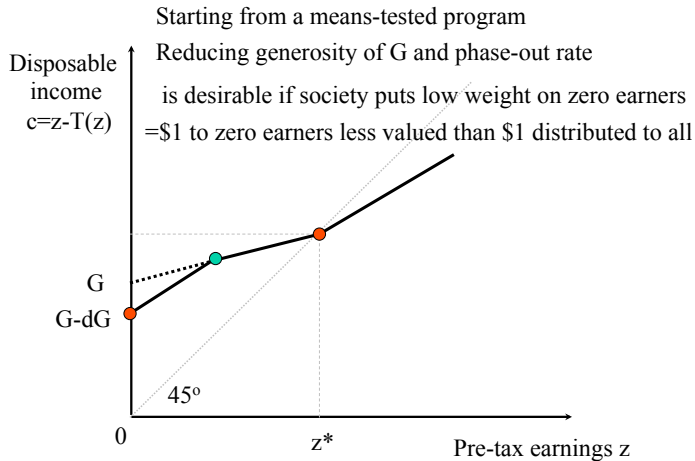
- If individuals respond to taxes only through intensive margin (how much they work at the margin and not whether they work), optimal transfer at bottom takes the form of a "negative income tax" or "universal basic income (UBI)" or "cash welfare":
 - Lumpsum grant $-T(0) > 0$ for those with no earnings
 - High MTRs $T'(z)$ at the bottom to phase-out the lumpsum grant quickly
- Intuition: high MTRs at bottom are efficient because:
 - (a) they target transfers to the most needy
 - (b) earnings at the bottom are low to start with $\Rightarrow$ intensive labor supply response does not generate large output losses

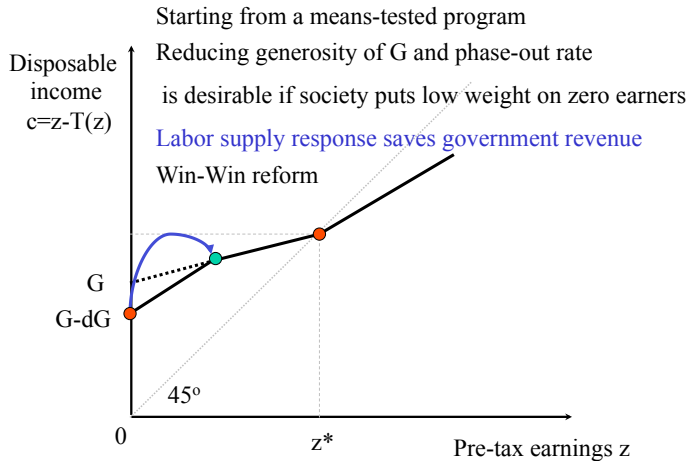
Optimal Profile of Transfers

- Universal basic income with high MTR at the bottom is all you need under the standard model
- But US system with zero MTR at the bottom justified if society sees people with zero income as less deserving than the average

Starting from a means-tested program

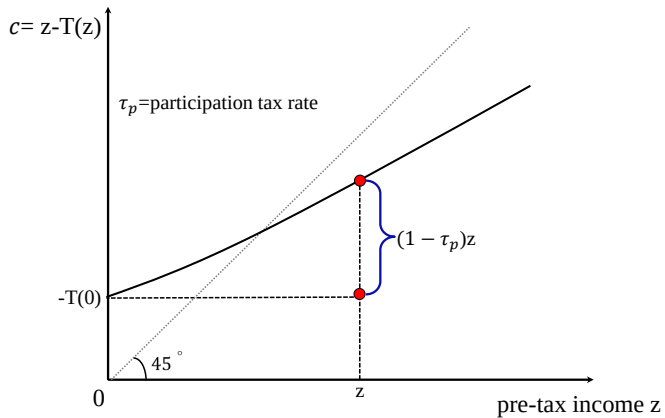




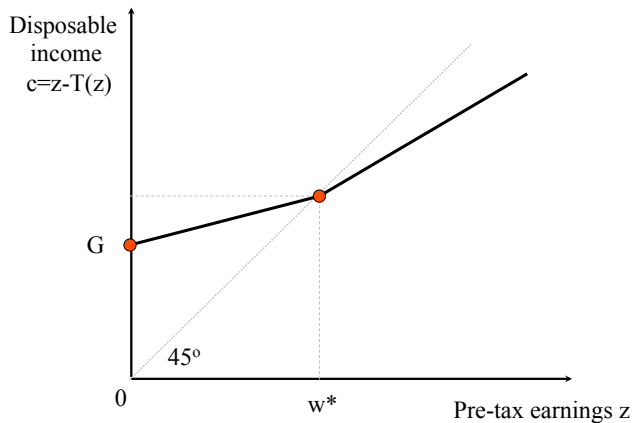


Optimal Transfers: Participation Responses

- Empirical literature shows that participation labor supply responses [whether to work or not] are large at the bottom [much larger and clearer than intensive responses]
- Participation depends on participation tax rate: $\tau_p = [T(z) - T(0)]/z$
- Individual keeps fraction $1 - \tau_p$ of earnings when moving from zero earnings to earnings z : $z - T(z) = -T(0) + z \cdot (1 - \tau_p)$
- **Key result:** in-work subsidies with $T'(z) < 0$ are optimal when labor supply responses are concentrated along extensive margin and govt cares about low income workers.



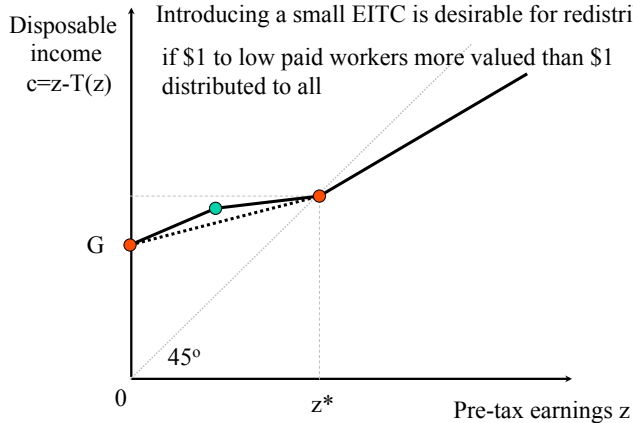
Starting from a Means-Tested Program



Starting from a Means-Tested Program

Introducing a small EITC is desirable for redistribution

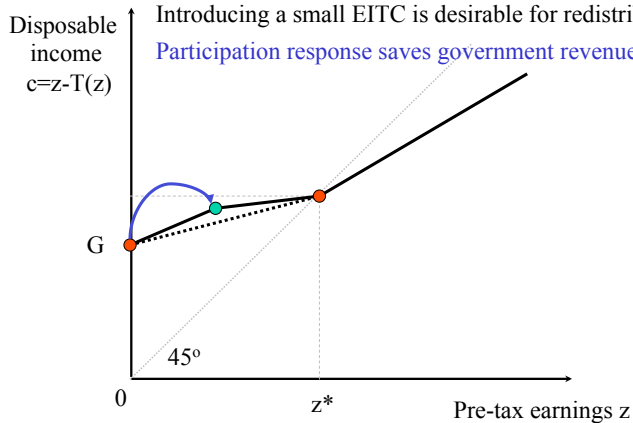
if \$1 to low paid workers more valued than \$1 distributed to all

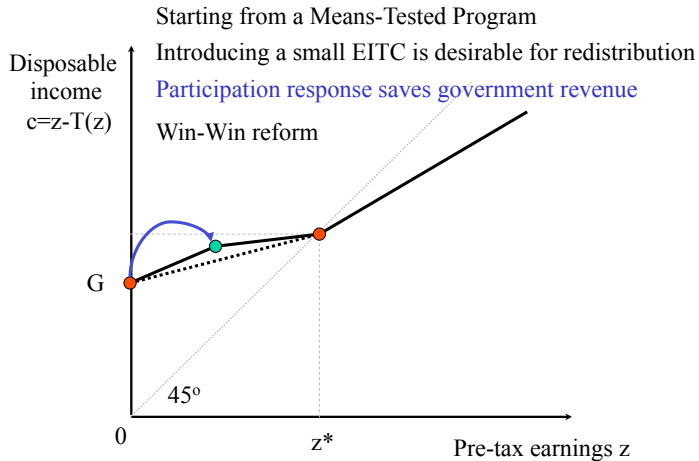


Starting from a Means-Tested Program

Introducing a small EITC is desirable for redistribution

Participation response saves government revenue



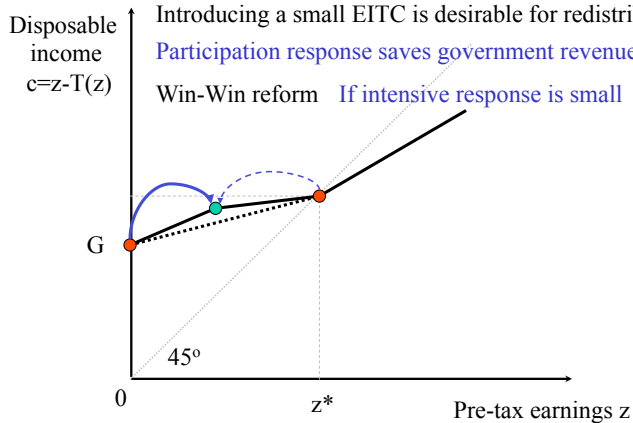


Starting from a Means-Tested Program

Introducing a small EITC is desirable for redistribution

Participation response saves government revenue

Win-Win reform If intensive response is small



Optimal Profile of Transfers: Summary

- ① Generous UBI with high MTR at bottom is optimal
- ② If society views **zero earners** as less deserving than average [e.g., view that substantial fraction of zero earners are “free loaders”] then lower UBI combined with low phasing out rate at bottom is optimal
- ③ If society views **low income workers** as more deserving than average [typically common view] and labor supply responses concentrated along extensive margin (work vs not) relative to the intensive margin (work a lot vs work a little) then low phasing out rate at bottom is optimal

- ① Means-tested transfer programs used to be of the traditional form with high phasing-out rates (sometimes above 100%)

⇒ No incentives to work (even with modest elasticities)
- ② Initially designed for groups not expected to work [widows in the US] but later attracting groups who could potentially work [single mothers]
- ③ In-work benefits have been introduced and expanded in OECD countries since 1980s (US EITC, UK Family Credit, etc.) and have been politically successful

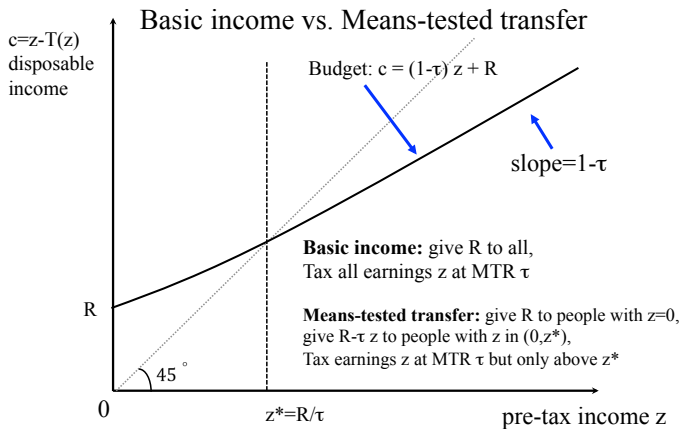
Basic Income vs. Means-tested transfer: Mankiw quiz

Consider an economy in which average income is \$50,000 but with much income inequality. To provide a social safety net, two possible policies are proposed.

- A. A universal transfer of \$10,000 to every person, financed by a 20-percent flat tax on income.
- B. A means-tested transfer of \$10,000. The full amount goes to someone without any income. The transfer is then phased out: You lose 20 cents of it for every dollar of income you earn. These transfers are financed by a tax of 20 percent on income above \$50,000.
- Which would you prefer?

Basic Income vs. Means-tested transfer

- Basic income definition: all people receive an unconditional sum of money (every year) regardless of how much they earn
- This is the R of the linear tax system $c = (1 - \tau) \cdot z + R$
- Or the $-T(0) > 0$ of the nonlinear tax system $c = z - T(z)$
- Basic income for everybody + higher taxes to fund it is **economically equivalent** to means-tested transfer phased out with earnings
- Pro basic income: less stigmatizing than means-tested transfer
- Cons: basic income requires higher “nominal” taxes (that are then rebated back)
- Countries provide “in-kind” basic income in the form of universal health care (not the US) and public education



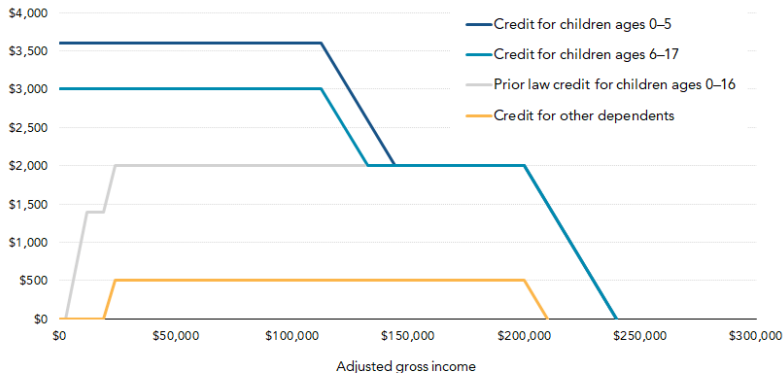
Modern social state and fairness

- Taxes roughly proportional to income = tax rate roughly equal = broad fairness appeal in isolation
- Social transfers (child care, education, health) roughly equal per person and de-commodified = everybody gets access to quality education and health care = broad fairness appeal
- Very redistributive combination but somewhat hidden $\Rightarrow$ Successful large social state (EU countries)
- Means-tested transfers are economically equivalent but reveal the redistribution $\Rightarrow$ Less generous social state (US)
- See Esping-Anderen '90 book for welfare-state regimes sociological analysis

Transfers or Tax Credits for Children

- Children reduce **normalized family income**
 - $\Rightarrow$ Children increase marginal utility of consumption
 - $\Rightarrow$ Transfer for children T_{kid} should be positive
- In practice, transfers for children are always positive
- Should $T_{kid}(z)$ increase with income z ?
 - Pro: rich spend more on their kids than lower income families
 - Cons: Lower income families need child transfers most
 - In practice, $T_{kid}(z)$ is fairly constant with z
 - Europe has much more generous pre-kindergarten child care benefits, US has more generous cash tax credits for families with children

FIGURE 1
Child Tax Credit, Single Parent
For one child, tax year 2021



Source: Urban-Brookings Tax Policy Center calculations.

Notes: Assumes all income comes from earnings, and child meets all tests to be a CTC-qualifying dependent. \$3,000 and \$3,600 credits are fully refundable; prior law limited refunds to \$1,400 out of the maximum \$2,000 credit. Credit for married parents first phases out at \$150,000 of income until credit reaches pre-2021 level; begins second phase out at \$400,000 of income. Only citizen children qualify for the \$3,000 and \$3,600 credits for children under 18. Noncitizens under age 18 who meet the dependency tests of eligibility can qualify for other dependent credit.

For both this lecture and the next lecture:

- Diamond, P. and E. Saez “From Basic Research to Policy Recommendations: The Case for a Progressive Tax”, Journal of Economic Perspectives, 25.4, (2011): 165-190. [\[web link\]](#)
- Mankiw, N.G., Weinzierl, M. and Yagan, D., 2009. Optimal taxation in theory and practice. The Journal of Economic Perspectives, 23(4), pp.147-174. [\[web link\]](#)
- Saez, Emmanuel, Joel Slemrod, and Seth H. Giertz. "The Elasticity of Taxable Income with Respect to Marginal Tax Rates: A Critical Review." Journal of Economic Literature 50(1) (2012): 3-50. [\[web link\]](#)

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- Diamond, P. and E. Saez “From Basic Research to Policy Recommendations: The Case for a Progressive Tax”, Journal of Economic Perspectives, 25.4, (2011): 165-190. [\[web link\]](#)
- IRS, Statistics of Income Division “U.S. Individual Income Tax: Personal Exemptions and Lowest and Highest Tax Bracket”(2013) [\[web link\]](#)

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- Saez, Emmanuel. "Optimal income transfer programs: intensive versus extensive labor supply responses." The Quarterly Journal of Economics 117.3 (2002): 1039-1073. [\[web link\]](#)
- Saez, Emmanuel, Joel Slemrod, and Seth H. Giertz. "The Elasticity of Taxable Income with Respect to Marginal Tax Rates: A Critical Review." Journal of Economic Literature 50(1) (2012): 3-50. [\[web link\]](#)
- Weinzierl, M. "The Promise of Positive Optimal Taxation: Normative Diversity and a Role for Equal Sacrifice." Journal of Public Economics 118 (2014): 128-142. [\[web link\]](#)